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|-----------------------------------------------------|----------|----------------|-------------------|
| Q.P Code                                            | D 112645 | Total Pages: 2 | Name              |
|                                                     |          |                | Register No.      |
| FIRST SEMESTER UG DEGREE EXAMINATION, NOVEMBER 2024 |          |                |                   |
| (CUFYUGP)                                           |          |                |                   |
| MAT1MN102 - CALCULUS OF A SINGLE VARIABLE           |          |                |                   |
| 2024 Admissions                                     |          |                |                   |
| Maximum Time :2 Hours                               |          |                | Maximum Marks :70 |

## Section A

All Question can be answered. Each Question carries 3 marks (Ceiling : 24 Marks)

|    |                                                                                                                                                                                                                                |
|----|--------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 1  | Does $\lim_{x \rightarrow 0^+} \frac{1}{x} = \lim_{x \rightarrow 0^-} \frac{1}{x}$ exist? Explain.                                                                                                                             |
| 2  | Find $\lim_{x \rightarrow 3^+} \frac{x}{x-3}$                                                                                                                                                                                  |
| 3  | Does the function $f(x) = \begin{cases} x^2 & \text{if } x \leq 2 \\ x+10 & \text{if } x > 2 \end{cases}$ , continuous at 2? Why?                                                                                              |
| 4  | During the first 30 s of a rocket flight, the rocket is propelled straight up so that in t seconds it reaches a height of $s = t^2 + 2$ ft. What is the average velocity of the rocket during the first 1000 ft of its flight? |
| 5  | Using the definition of differentiation find $\frac{d}{dx}[3x]$                                                                                                                                                                |
| 6  | Find $\frac{dy}{dx}$ if $y = \log_{25}(x^2 - 1)$                                                                                                                                                                               |
| 7  | Find $\lim_{x \rightarrow 0^+} \frac{\tan x}{x}$ , $x \in (0, \pi/2)$                                                                                                                                                          |
| 8  | Use logarithmic differentiation to find $\frac{d}{dx}[2^{\cos x + \ln x}]$                                                                                                                                                     |
| 9  | Explain a increasing function with an example                                                                                                                                                                                  |
| 10 | Find all critical points of $f(x) = x^3 - 3x + 1$                                                                                                                                                                              |

## Section B

All Question can be answered. Each Question carries 6 marks (Ceiling : 36 Marks))

|    |                                                                                                                                                                                                     |
|----|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 11 | Find $\lim_{x \rightarrow +\infty} \frac{e^x + e^{-x}}{e^x - e^{-x}}$                                                                                                                               |
| 12 | Show that $ x $ is continuous everywhere                                                                                                                                                            |
| 13 | Let $f(x) = \begin{cases} x^2 & \text{if } x \leq 1 \\ 4\sqrt{x} & \text{if } x > 1 \end{cases}$ Determine whether $f$ is differentiable at $x = 1$ . If so, find the value of the derivative there |
| 14 | Find $\frac{dy}{dx}$ , if $y = \sin(\sqrt{1 + \tan x})$                                                                                                                                             |
| 15 | Use implicit differentiation to find all points on the graph of $y^4 + y^2 = x(x - 1)$ at which the tangent line is vertical.                                                                       |
| 16 | Evaluate $\lim_{x \rightarrow 0^+} x^3 \ln x$ ; $x \in (0, \infty)$                                                                                                                                 |
| 17 | Find the inflection points of the function $f(x) = x^3 - 3x^2 + 1$                                                                                                                                  |
| 18 | Find the relative extrema of $f(x) = \frac{x^2}{x^4 + 16}$                                                                                                                                          |

## Section C

Answer any ONE. Each Question carries 10 marks (1x10=10 Marks))

|    |                                                                                                                                                                                                                                                                                                                   |
|----|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| 19 | On a sunny day, a 50 ft flagpole casts a shadow that changes with the angle of elevation of the Sun. Let $s$ be the length of the shadow and $\theta$ the angle of elevation of the Sun. Find the rate at which the length of the shadow is changing with respect to $\theta$ when $\theta = 45^\circ$ .          |
| 20 | For the function $f(x) = x^4 - 5x^3 + 9x^2$ <ol style="list-style-type: none"> <li>Find the intervals on which <math>f</math> is increasing and decreasing</li> <li>Find the open intervals on which <math>f</math> is concave up and concave down</li> <li>Find all of its inflection points, if any.</li> </ol> |